

JOSHUA HIMMENS

Engineering Physics Student

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WORK EXPERIENCE

AltaML

Associate Machine Learning Developer

Jan 2025 – Apr 2025

TRIUMF

ATLAS Deep Learning Research Student

May 2024 – Feb 2025

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with Wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting edge transfer learning approaches.
- Implementing models into production using NVIDIA Triton.

UBC Orbit Satellite Team

Command and Data Handling (CDH) Lead and Firmware Developer

Oct 2023 – Present

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of 10 firmware developers, with over 40 tasks completed, 1300 CI builds, 2000 lines of code completed/written.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

Embedded Firmware Developer

Jan 2023 – Present

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computer for launch in Q1 2026.

NumPy

Open Source Contributor

Jan 2021 – Dec 2021

- Contributed to the open-source numerical analysis tool with 86 million monthly downloads.

Elections Alberta

Elections Officer

Jan 2023 – Dec 2023

- Integrated as part of a team to run a safe and fair provincial election.
- Managed public interactions and vote counting.

Western Canada High School

International Baccalaureate Student

Jan 2020 – Dec 2023

- Took the IB diploma program with higher-level Math, Physics, and Chemistry.
- Completed an Extended Essay on simulations of the brachistochrone problem in physics.

HONORS AND AWARDS

Hector J. MacLeod Award (\$2000)

Issued by IEEE Vancouver Section

Apr 2024

- Awarded to students who have demonstrated pioneering efforts in their chosen field of study and attained high scholastic honours.

Erich Vogt First Year Summer Research Experience (FYSRE) Award

Issued by UBC Department of Physics and Astronomy

May 2024

- Awarded to promising students in physics for a 4-month research placement.

Alberta Centennial Award and Alberta Premier's Citizenship Award (\$2000)

Issued by Government of Alberta

May 2023

- Awarded for outstanding community service.

Calgary Flames Foundation Community Involvement Scholarship (\$2005)

Issued by Education Matters

Aug 2023

- Awarded for community involvement.

Julia Turnbull Leadership Award for exceptional community service (\$1000)

Issued by *The Calgary Foundation*

Jul 2023

- Awarded for exceptional community service.

Ted Rogers Entrance Scholarship (\$2000)

Issued by *Rogers Communications*

Jun 2023

- Awarded for academic achievement.

Tom Lawson Award for embodying the spirit of Canadian debate

Issued by *Canadian National Debate Society*

Jan 2023

- Required a vote of the student delegates at the Canadian National Debate Seminar.

10 Scouts Canada commendations

Issued by *Scouts Canada*

May 2022

- Awarded for various achievements.